Smart Library Request Workflow in ServiceNow

Detailed Project Summary Report

Platform : ServiceNow

Project Domain: Library services

Abstract

The Smart Library Request Workflow is an enterprise application built on the ServiceNow platform to automate library service requests such as book borrowing, journal access, and resource allocation. Traditional library request handling through paper forms or emails often leads to delays, duplication, and lack of transparency. By leveraging ServiceNow's Flow Designer, Business Rules, Notifications, and Dashboards, this workflow centralizes request management, ensures faster approvals, and provides real-time visibility.

1. Introduction

1.1 Background

Libraries in academic and corporate environments handle thousands of requests daily. Manual processes often result in:

- Delayed approvals
- Duplicate requests

- Inefficient tracking

1.2 Purpose

The Smart Library Workflow aims to:

- Automate request submission and approvals
- Provide dashboards for administrators
- Ensure secure access to library resources
- Improve user experience for students, faculty, and staff

2. Ideation Phase

2.1 Problem Statement

- Librarians struggle with manual request tracking.
- Students face delays in accessing resources.
- Administrators lack visibility into request volumes.

2.2 Empathy Map

- **Think & Feel:** Concerned about delays and fairness.
- **Hear:** Complaints from students about slow approvals.
- **See:** Piles of request forms and emails.
- **Say & Do:** Follow up manually with departments.
- **Pain:** Duplication and inefficiency.
- **Gain:** Automated workflows and dashboards.

2.3 Idea Prioritization

Options considered:

- Spreadsheet-based management
- Standalone portal
- ERP integration
- **ServiceNow-based workflow (chosen for scalability and automation)**

3. Requirement Analysis

3.1 Functional Requirements

FR No	Description
FR1	Request creation
FR2	Resource allocation
FR3	Approval workflow
FR4	Notifications
FR5	Dashboards & reports

3.2 Non-Functional Requirements

- Usability: Simple forms
- Security: Role-based access control
- Reliability: Prevent duplicate requests
- Scalability: Handle thousands of requests

- Availability: Cloud-based 24/7

3.3 Technology Stack

- **Platform:** ServiceNow
- **UI:** Forms & Service Portal
- **Logic:** Flow Designer & Business Rules
- **Database:** ServiceNow Tables
- **Automation:** Notifications & SLA
- **Reporting:** Dashboards

4. Project Development Phase

- Instance setup
- Request table creation (Book, Journal, Resource)
- Workflow automation for approvals
- Notification setup
- Dashboard creation
- Security roles & ACLs

5. Sprint-by-Sprint Activities

Sprint 1 – Instance Setup

- Activity: Create ServiceNow developer account
- Deliverable: Active instance with admin access
- Outcome: Library team begins configuration

Sprint 2 – Data Modeling

Table	Key Fields	Relationships
Request	Title, User, Status	Linked to Resource & User
Resource	Name, Type, Availability	Linked to Request
User	Name, Email, Role	Linked to Request

Sprint 3 – Workflow Automation

Workflow:
[Request Submitted]

→ [Librarian Review]

→ [Department Head Approval]

→ [Resource Allocation]

→ [Notification Sent]

Sprint 4 – Approvals

Approval chain: Librarian → Department Head → Administrator

Sprint 5 – Reporting

Dashboards:

- Pending vs approved requests
- Resource utilization charts

Sprint 6 – Security

Role hierarchy:

- Administrator
- Librarian

- Student/Faculty (limited access)

6. Testing Strategies

- **Unit Testing:** Validate request table fields.
- **Integration Testing:** Ensure workflows trigger correctly.
- **System Testing:** Bulk request simulation (e.g., 500 requests).
- **UAT:** Library staff test workflows in real scenarios.

7. Advantages & Disadvantages

Advantages

- Centralized request management
- Automated workflows
- Real-time reporting
- Improved efficiency

Disadvantages

- Initial configuration effort
- Dependency on ServiceNow

8. Compliance & Risk Management

Key Considerations

- Confidentiality: Requests accessible only to authorized staff
- Integrity: Validation rules prevent duplication

- Availability: Cloud-based 24/7 access
- Auditability: Logs maintained for every workflow

Risks & Mitigation

- **Risk:** Duplicate requests → **Mitigation:** Validation rules
- **Risk:** Workflow misconfiguration → **Mitigation:** Testing before deployment
- **Risk:** Unauthorized access → **Mitigation:** Strict ACLs

9. Case Studies

Case Study 1 – Request Efficiency

- Problem: Manual approvals took 3 days
- Solution: Automated workflow
- Outcome: Reduced to 1 day

Case Study 2 – Duplication Control

- Problem: Duplicate requests common
- Solution: Validation rules
- Outcome: 90% reduction

Case Study 3 – Compliance Reporting

- Problem: Reports took hours
- Solution: Dashboards
- Outcome: Reports in minutes

10. Performance Analysis

- **Response Time:** Dashboard refresh < 2s, request search < 1s
- **Scalability:** Tested with 5,000 requests
- **Reliability:** No duplicate records, consistent workflows

11. Validation Methods

- **Data Validation:** Mandatory fields enforced
- **Workflow Validation:** Approval chains follow hierarchy
- **Security Validation:** ACLs tested for restricted access

12. Explanation Section

ServiceNow enables modular workflow automation. The Smart Library Workflow replaces manual request handling with automated processes, improving transparency, reducing errors, and ensuring compliance.

13. Future Enhancements

- AI-driven analytics for resource demand prediction
- Chatbot integration for self-service requests
- Mobile app extension for request submission and approvals
- ERP integration for finance and resource tracking